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S.No. of Question Paper : 2815

Unique Paper Code : 32351402

GC-4

Name of the Paper : C-9, Riemann Integration and Series of Functions

Name of the Course : B.Sc. (H) Mathematics

Semester : IV

Duration : 3 Hours

Maximum Marks : 75

(Write your Roll No. on the top immediately on receipt of this question paper.)

Attempt any two parts from each question.

All questions are compulsory.

1. (a) Prove that a bounded function f on $[a, b]$ is integrable if and only if for each $\varepsilon > 0$, $\exists \delta > 0$ such that :

$$\text{mesh}(P) < \delta \Rightarrow U(f, P) - L(f, P) < \varepsilon,$$

$\forall$ partitions P of $[a, b]$. 6

- (b) Show that every continuous function on $[a, b]$ is integrable. Is the converse true ? Justify. 6

- (c) State and prove Intermediate Value Theorem for integrals. Show, with the help of an example, that the condition of continuity cannot be relaxed. 6

2. (a) (i) Let f be a bounded function on $[a, b]$. Define the Darboux Integral $\int_a^b f$.

(ii) Give an example of a function f which is not integrable, for which $|f|$ is integrable.

(b) (i) State Fundamental Theorem of Calculus-II.

(ii) Let f be a continuous function on $\mathbf{R}$, and define :

$$F(x) = \int_{x-1}^{x+1} f(t) dt \text{ for } x \in \mathbf{R}$$

Show that F is differentiable on $\mathbf{R}$, and compute F' .

(c) (i) Define :

$$f(x) = [x] \text{ on } [0, 3].$$

Show that f is integrable, and evaluate

$$\int_0^3 f(x) dx.$$

(ii) Define :

$$g : [0, 2] \rightarrow [0, 4] \text{ as } g(x) = x^2.$$

Let :

$$P = \{0, \frac{1}{2}, 1, \frac{3}{2}, 2\}.$$

Find $U(g, P)$.

(a) Show that :

$$\int_0^{\infty} e^{-t} t^{x-1} dt$$

converges $\Leftrightarrow x > 0$.

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(b) Determine the convergence or divergence of :

$$\int_0^{\infty} \frac{1}{e^x \sqrt{x}} dx$$

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(c) In the following, determine the values of r for which the integral converges/diverges. Determine the values of integrals in case of convergence.

(i) $\int_1^{\infty} x^{-r} dx$

(ii) $\int_0^{\infty} x^{-r} dx$

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(a) Let $\{f_n\}$ be a sequence of integrable functions on $[a, b]$ and suppose that $\{f_n\}$ converges uniformly on $[a, b]$ to f . Show that f is integrable. 6

(b) Let

$$f_n = \frac{1}{(1+x)^n}$$

for x in $[0, 1]$. Find the point-wise limit f of the sequence $\{f_n\}$ on $[0, 1]$.

Does $\{f_n\}$ converge uniformly to f on $[0, 1]$? Justify. 6

P.T.O.

- (c) Show that if $0 < b < 1$, then the convergence of the sequence :

$$f_n = \frac{x^n}{(1+x^n)},$$

for x in $\mathbb{R}, x \geq 0$ is uniform on the interval $[0, b]$, but is not uniform on the interval $[0, 1]$. 6

5. (a) State and prove Weierstrass M-Test and hence examine the uniform convergence of the series :

$$\sum_{n=1}^{\infty} \frac{1}{(x^2 + n^2)}, x \text{ in } \mathbb{R}.$$

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- (b) Show that the series of functions :

$$\sum \frac{1}{(1+x^n)}, x \neq 0,$$

converges uniformly on $[a, \infty)$ for $a > 1$, but is not uniformly convergent on $(1, \infty)$. 6

- (c) Let $f: A \rightarrow \mathbb{R}, A \subseteq \mathbb{R}$ be a bounded function on A . Define uniform norm $\|f\|$ on A . State criterion for uniform convergence of a sequence of bounded functions using uniform norm and hence examine for uniform convergence the sequence :-

$$f_n = x^n; x \in [0, 1].$$

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(a) (i) Given :

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$$

for $|x| < 1$. Evaluate $\sum_{n=1}^{\infty} \frac{n^2}{2^n}$. 2.5

(ii) Let

$$f(x) = \sum a_n x^n$$

for $|x| < R$. If $f(x) = f(-x)$, $\forall |x| < R$, then show that $a_n = 0$ for all odd values of n . 4

(b) (i) If the power series $\sum_{n=0}^{\infty} a_n x^n$ has radius of convergence R , then show that series :

$$\sum_{n=1}^{\infty} n a_n x^{n-1} \quad \text{and} \quad \sum_{n=0}^{\infty} \frac{a_n x^{n+1}}{n+1}$$

also have the radius of convergence R . 4

(ii) Show that :

$$f(x) = \sum_{n=0}^{\infty} \frac{x^n}{n!} \quad \text{for } x \in \mathbb{R}$$

is well-defined. Also show that :

$$f(x) = f'(x), \quad \text{for } x \in \mathbb{R}.$$

2.5

P.T.O.

- (c) (i) Define interval of convergence for a power series.
Find the radius of convergence and interval of convergence for the following power series :

$$\sum_{n=1}^{\infty} \frac{x^n}{n}, \quad \sum_{n=1}^{\infty} \frac{x^n}{n^2}.$$

- (ii) State Weierstrass's Approximation Theorem.